

# FMv3 Analysis: Continuous-t Velocity Field

Three-way comparison — A: Diffusion S6 · F: FMv2 ODE=10 · G: FMv3 continuous-t (ODE=10)

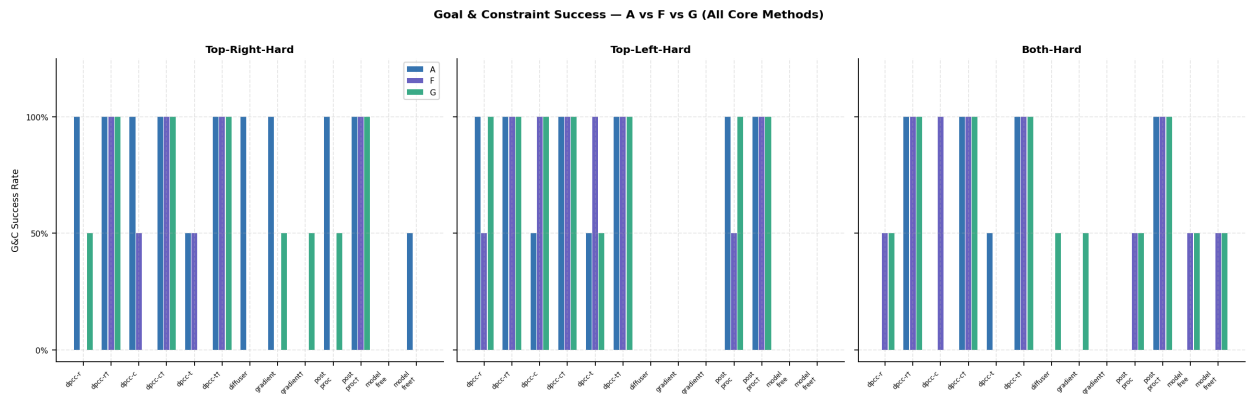
|                             |                                   |                                            |
|-----------------------------|-----------------------------------|--------------------------------------------|
| A = Diffusion S6 · baseline | F = FMv2 ODE=10 · discrete-t bins | G = FMv3 ODE=10 · continuous real-valued t |
|-----------------------------|-----------------------------------|--------------------------------------------|

**Key architectural change in G (FMv3):** the velocity field is queried with a continuous real-valued time  $t \in [0,1]$  (SafeFlow-style) instead of quantizing  $t$  into discrete bins before model evaluation as in F (FMv2). All other settings are identical: same action\_weight=1, ODE steps=10, model checkpoint step 90000, Seed 6. Comparing G vs F isolates the pure effect of continuous-t conditioning on trajectory quality, constraint satisfaction, and computational cost.

# Section 1 — Visual Analysis: A vs F vs G

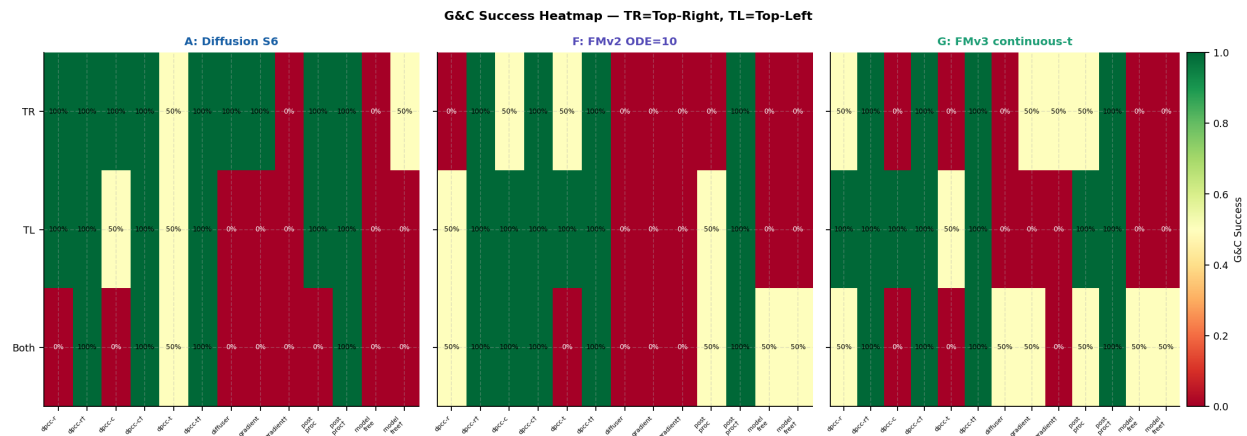
## 1.1 Goal & Constraint Success — All Core Methods

Primary metric. Blue=A (Diffusion), purple=F (FMv2 discrete-t), green=G (FMv3 continuous-t). †=tightened.



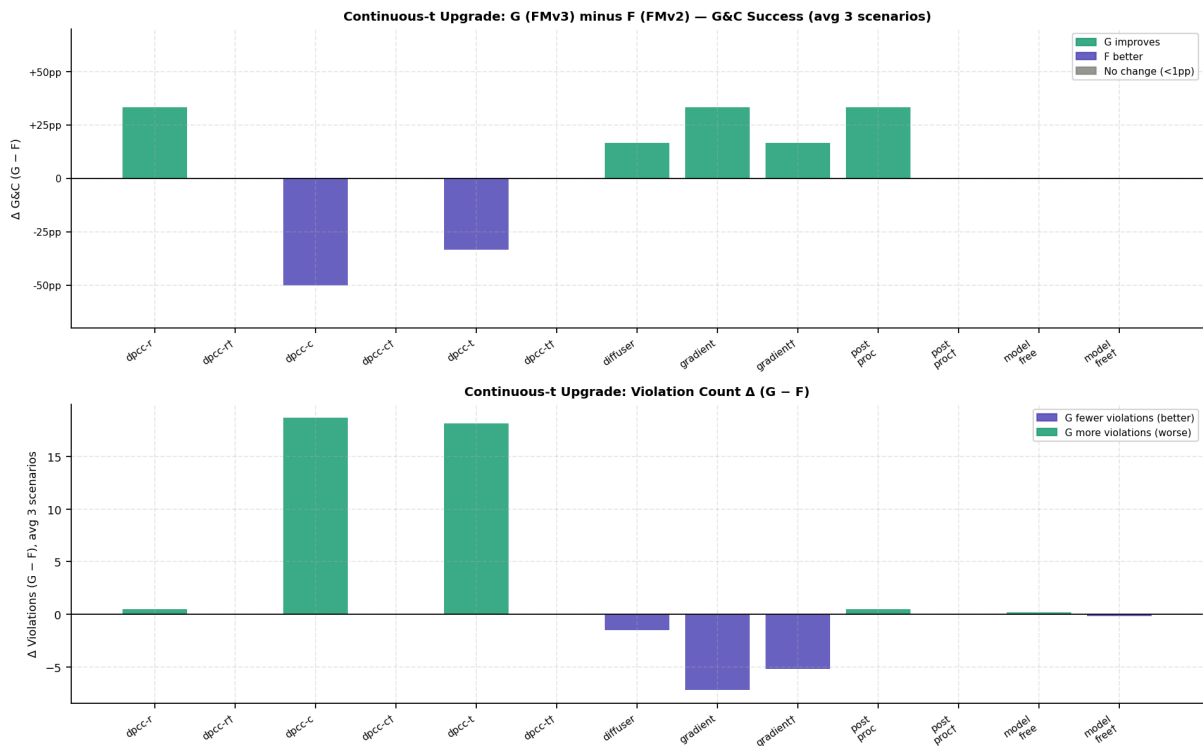
## 1.2 G&C; Heatmap — All 3 Configurations

Green=100%, yellow=50%, red=0%. TR=Top-Right, TL=Top-Left.

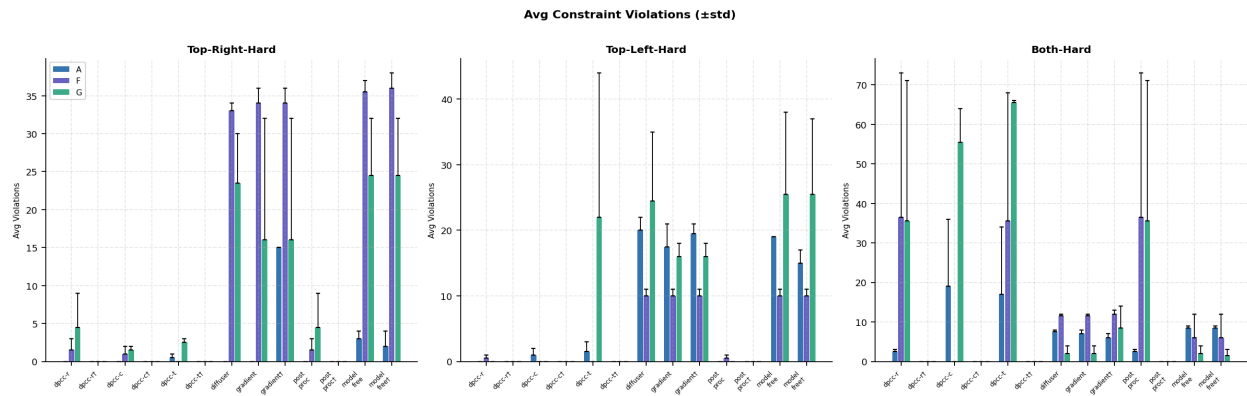


### 1.3 Continuous-t Upgrade Impact: G minus F

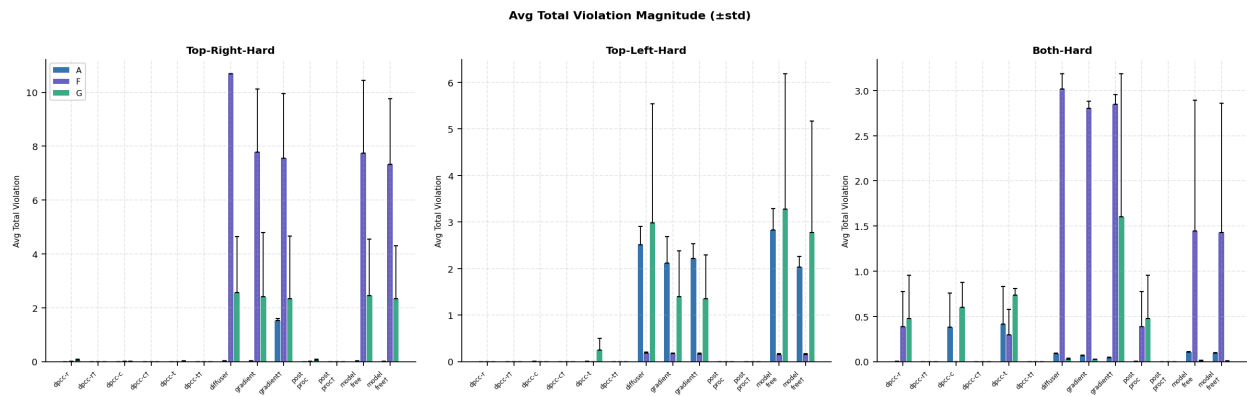
Top: G&C; success change from switching discrete-t  $\rightarrow$  continuous-t (avg 3 scenarios). Green bars = G improves, purple bars = F was better, gray = no meaningful change. Bottom: violation count delta — purple = G reduces violations (better).



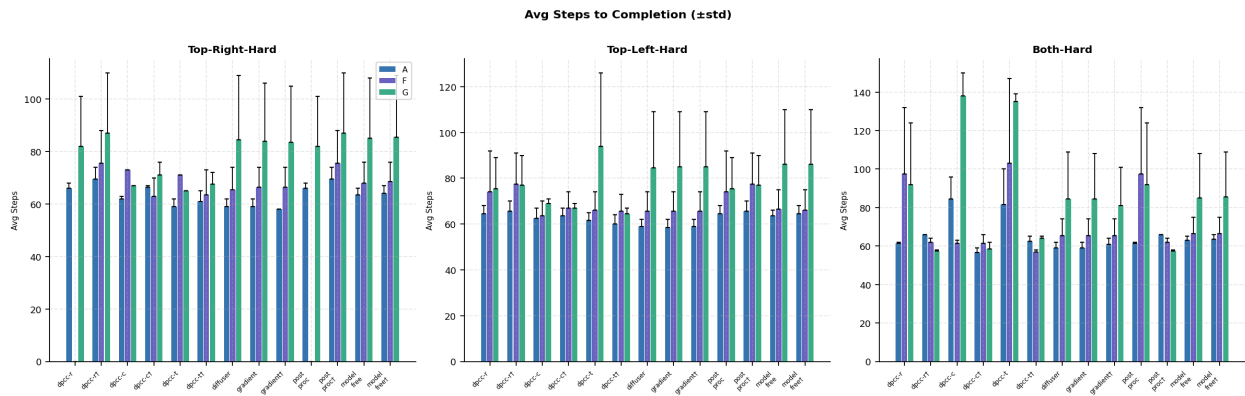
## 1.4 Avg Constraint Violations ( $\pm$ std)



## 1.5 Avg Total Violation Magnitude ( $\pm$ std)

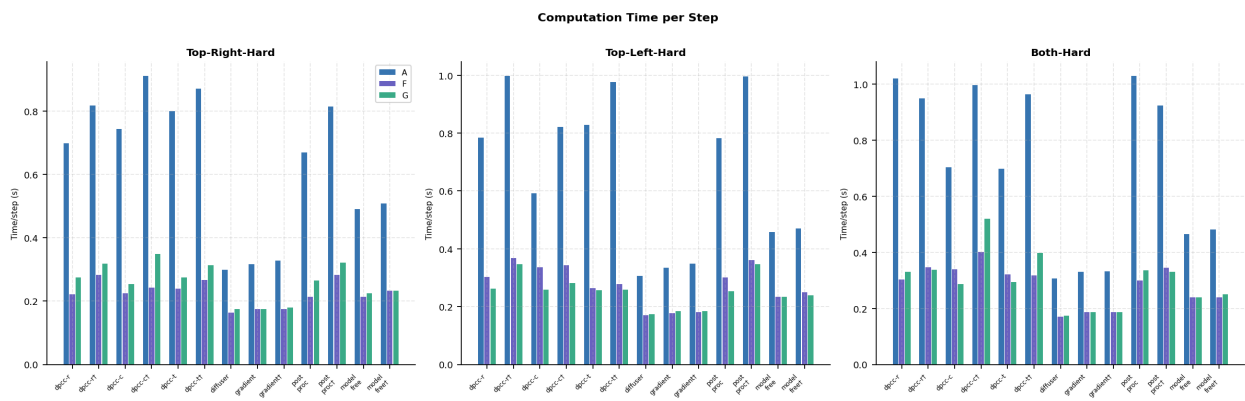


## 1.6 Avg Steps to Completion ( $\pm$ std)



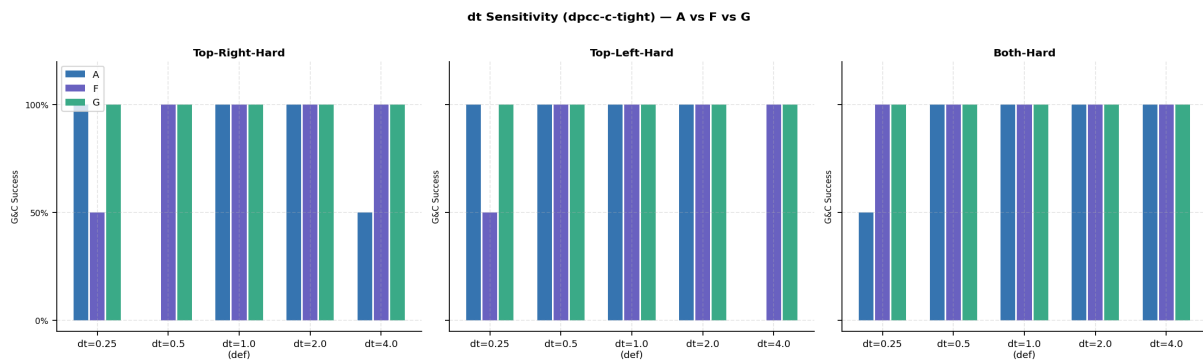
## 1.7 Computation Time per Step

*G uses continuous- $t$  ODE integration — compute cost comparison vs discrete- $t$  F and Diffusion A.*

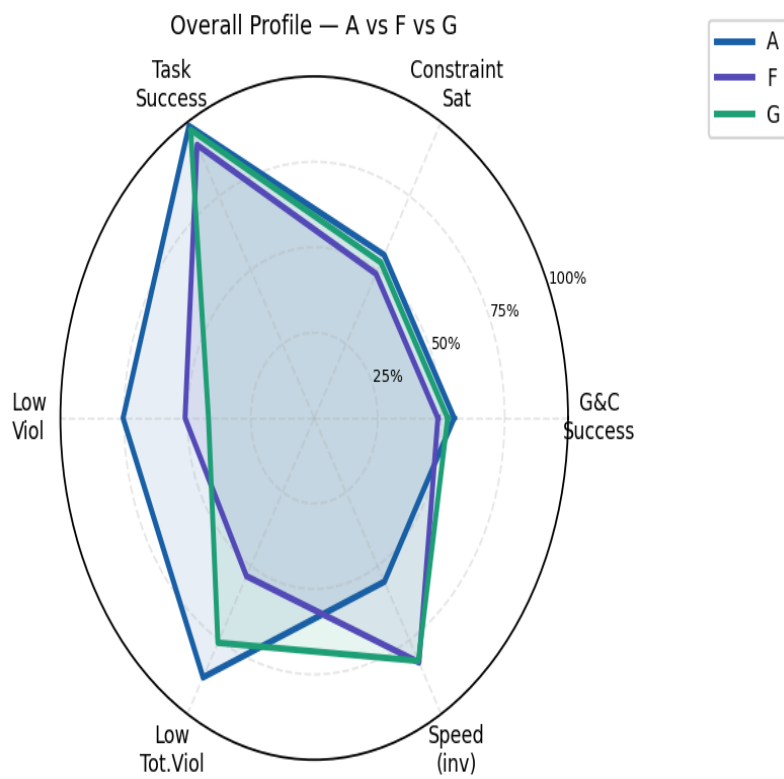


## 1.8 dt Sensitivity — dpcc-c-tightened: A vs F vs G

G&C; success across  $dt \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$ . G resolves F's  $dt=0.25$  instability on top-left ( $137 \pm 37$  steps, 51 violations  $\rightarrow$  clean 76 steps, 0 violations).



## 1.9 Overall Model Profile Radar



## Section 2 — Complete Raw Data Tables

All metrics for G (FMv3), F (FMv2), and A (Diffusion). **SR**=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps±std**, **Viol#±std**, **ViolΣ±std**, **Time(s/step)**, **TE**=Tracking Error.

### 2.1 Top-Right-Hard

#### G: FMv3 continuous-t — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std | Viol#±std | ViolΣ±std   | Time  | TE    |
|---------------------|------|------|------|-----------|-----------|-------------|-------|-------|
| dpcc-r              | 100% | 50%  | 50%  | 82.0±19.0 | 4.5±4.5   | 0.045±0.045 | 0.275 | —     |
| dpcc-r-tight        | 100% | 100% | 100% | 87.0±23.0 | 0.0±0.0   | 0.000±0.000 | 0.318 | —     |
| dpcc-c              | 50%  | 0%   | 0%   | 67.0±0.0  | 1.5±0.5   | 0.005±0.004 | 0.253 | —     |
| dpcc-c-tight        | 100% | 100% | 100% | 71.0±5.0  | 0.0±0.0   | 0.000±0.000 | 0.349 | —     |
| dpcc-t              | 50%  | 0%   | 0%   | 65.0±0.0  | 2.5±0.5   | 0.018±0.016 | 0.275 | —     |
| dpcc-t-tight        | 100% | 100% | 100% | 67.5±4.5  | 0.0±0.0   | 0.000±0.000 | 0.314 | —     |
| diffuser            | 100% | 0%   | 0%   | 84.5±24.5 | 23.5±6.5  | 2.563±2.077 | 0.175 | 0.029 |
| gradient            | 100% | 50%  | 50%  | 84.0±22.0 | 16.0±16.0 | 2.405±2.386 | 0.175 | —     |
| gradient-tight      | 100% | 50%  | 50%  | 83.5±21.5 | 16.0±16.0 | 2.337±2.318 | 0.180 | —     |
| post_proc           | 100% | 50%  | 50%  | 82.0±19.0 | 4.5±4.5   | 0.045±0.045 | 0.265 | —     |
| post_proc-tight     | 100% | 100% | 100% | 87.0±23.0 | 0.0±0.0   | 0.000±0.000 | 0.321 | —     |
| model_free          | 100% | 0%   | 0%   | 85.0±23.0 | 24.5±7.5  | 2.453±2.092 | 0.224 | —     |
| model_free-tight    | 100% | 0%   | 0%   | 85.5±23.5 | 24.5±7.5  | 2.334±1.967 | 0.232 | —     |
| dpcc-c-tight-dt0.25 | 100% | 100% | 100% | 78.5±3.5  | 0.0±0.0   | 0.002±0.000 | 0.681 | —     |
| dpcc-c-tight-dt0.5  | 100% | 100% | 100% | 77.5±5.5  | 0.0±0.0   | 0.001±0.000 | 0.426 | —     |
| dpcc-c-tight-dt2.0  | 100% | 100% | 100% | 71.0±6.0  | 0.0±0.0   | 0.000±0.000 | 0.272 | —     |
| dpcc-c-tight-dt4.0  | 100% | 100% | 100% | 131.5±1.5 | 0.0±0.0   | 0.000±0.000 | 0.270 | —     |

#### F: FMv2 ODE=10 discrete-t — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std | Viol#±std | ViolΣ±std    | Time  | TE    |
|---------------------|------|------|------|-----------|-----------|--------------|-------|-------|
| dpcc-r              | 0%   | 0%   | 0%   | 0.0±0.0   | 1.5±1.5   | 0.008±0.008  | 0.222 | —     |
| dpcc-r-tight        | 100% | 100% | 100% | 75.5±12.5 | 0.0±0.0   | 0.000±0.000  | 0.283 | —     |
| dpcc-c              | 50%  | 50%  | 50%  | 73.0±0.0  | 1.0±1.0   | 0.001±0.001  | 0.224 | —     |
| dpcc-c-tight        | 100% | 100% | 100% | 63.0±7.0  | 0.0±0.0   | 0.000±0.000  | 0.243 | —     |
| dpcc-t              | 50%  | 50%  | 50%  | 71.0±0.0  | 0.0±0.0   | 0.000±0.000  | 0.239 | —     |
| dpcc-t-tight        | 100% | 100% | 100% | 63.5±9.5  | 0.0±0.0   | 0.000±0.000  | 0.267 | —     |
| diffuser            | 100% | 0%   | 0%   | 65.5±8.5  | 33.0±1.0  | 10.682±0.010 | 0.164 | 0.029 |
| gradient            | 100% | 0%   | 0%   | 66.5±7.5  | 34.0±2.0  | 7.776±2.344  | 0.175 | —     |
| gradient-tight      | 100% | 0%   | 0%   | 66.5±7.5  | 34.0±2.0  | 7.560±2.399  | 0.175 | —     |
| post_proc           | 0%   | 0%   | 0%   | 0.0±0.0   | 1.5±1.5   | 0.008±0.008  | 0.213 | —     |
| post_proc-tight     | 100% | 100% | 100% | 75.5±12.5 | 0.0±0.0   | 0.000±0.000  | 0.283 | —     |
| model_free          | 100% | 0%   | 0%   | 68.0±8.0  | 35.5±1.5  | 7.743±2.693  | 0.214 | —     |
| model_free-tight    | 100% | 0%   | 0%   | 68.5±7.5  | 36.0±2.0  | 7.332±2.428  | 0.233 | —     |
| dpcc-c-tight-dt0.25 | 100% | 50%  | 50%  | 90.5±12.5 | 1.0±1.0   | 0.003±0.003  | 0.452 | —     |

|                    |      |      |      |           |         |             |       |   |
|--------------------|------|------|------|-----------|---------|-------------|-------|---|
| dpcc-c-tight-dt0.5 | 100% | 100% | 100% | 72.0±11.0 | 0.0±0.0 | 0.000±0.000 | 0.275 | – |
| dpcc-c-tight-dt2.0 | 100% | 100% | 100% | 72.5±0.5  | 0.0±0.0 | 0.000±0.000 | 0.248 | – |
| dpcc-c-tight-dt4.0 | 100% | 100% | 100% | 142.0±4.0 | 0.0±0.0 | 0.000±0.000 | 0.260 | – |

A: Diffusion — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std | Viol#±std | Viol $\Sigma$ ±std | Time  | TE    |
|---------------------|------|------|------|-----------|-----------|--------------------|-------|-------|
| dpcc-r              | 100% | 100% | 100% | 66.0±2.0  | 0.0±0.0   | 0.000±0.000        | 0.698 | –     |
| dpcc-r-tight        | 100% | 100% | 100% | 69.5±4.5  | 0.0±0.0   | 0.000±0.000        | 0.817 | –     |
| dpcc-c              | 100% | 100% | 100% | 62.0±1.0  | 0.0±0.0   | 0.000±0.000        | 0.744 | –     |
| dpcc-c-tight        | 100% | 100% | 100% | 66.5±0.5  | 0.0±0.0   | 0.000±0.000        | 0.911 | –     |
| dpcc-t              | 100% | 50%  | 50%  | 59.0±3.0  | 0.5±0.5   | 0.000±0.000        | 0.799 | –     |
| dpcc-t-tight        | 100% | 100% | 100% | 61.0±4.0  | 0.0±0.0   | 0.000±0.000        | 0.871 | –     |
| diffuser            | 100% | 100% | 100% | 59.0±3.0  | 0.0±0.0   | 0.021±0.002        | 0.298 | 0.025 |
| gradient            | 100% | 100% | 100% | 59.0±3.0  | 0.0±0.0   | 0.021±0.002        | 0.316 | –     |
| gradient-tight      | 50%  | 0%   | 0%   | 58.0±0.0  | 15.0±0.0  | 1.521±0.082        | 0.327 | –     |
| post_proc           | 100% | 100% | 100% | 66.0±2.0  | 0.0±0.0   | 0.000±0.000        | 0.670 | –     |
| post_proc-tight     | 100% | 100% | 100% | 69.5±4.5  | 0.0±0.0   | 0.000±0.000        | 0.814 | –     |
| model_free          | 100% | 0%   | 0%   | 63.5±2.5  | 3.0±1.0   | 0.012±0.011        | 0.490 | –     |
| model_free-tight    | 100% | 50%  | 50%  | 64.0±3.0  | 2.0±2.0   | 0.009±0.009        | 0.508 | –     |
| dpcc-c-tight-dt0.25 | 100% | 100% | 100% | 81.0±6.0  | 0.0±0.0   | 0.002±0.000        | 2.225 | –     |
| dpcc-c-tight-dt0.5  | 100% | 0%   | 0%   | 75.5±2.5  | 1.0±0.0   | 0.003±0.000        | 1.668 | –     |
| dpcc-c-tight-dt2.0  | 100% | 100% | 100% | 66.5±0.5  | 0.0±0.0   | 0.000±0.000        | 0.689 | –     |
| dpcc-c-tight-dt4.0  | 50%  | 50%  | 50%  | 72.0±0.0  | 0.0±0.0   | 0.000±0.000        | 0.535 | –     |

## 2.2 Top-Left-Hard

G: FMv3 continuous-t — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std | Viol#±std | Viol $\Sigma$ ±std | Time  | TE    |
|---------------------|------|------|------|-----------|-----------|--------------------|-------|-------|
| dpcc-r              | 100% | 100% | 100% | 75.5±13.5 | 0.0±0.0   | 0.000±0.000        | 0.262 | –     |
| dpcc-r-tight        | 100% | 100% | 100% | 77.0±13.0 | 0.0±0.0   | 0.000±0.000        | 0.346 | –     |
| dpcc-c              | 100% | 100% | 100% | 69.0±2.0  | 0.0±0.0   | 0.000±0.000        | 0.258 | –     |
| dpcc-c-tight        | 100% | 100% | 100% | 67.0±2.0  | 0.0±0.0   | 0.000±0.000        | 0.282 | –     |
| dpcc-t              | 100% | 50%  | 50%  | 94.0±32.0 | 22.0±22.0 | 0.250±0.250        | 0.257 | –     |
| dpcc-t-tight        | 100% | 100% | 100% | 64.5±2.5  | 0.0±0.0   | 0.000±0.000        | 0.259 | –     |
| diffuser            | 100% | 0%   | 0%   | 84.5±24.5 | 24.5±10.5 | 2.978±2.559        | 0.173 | 0.029 |
| gradient            | 100% | 0%   | 0%   | 85.0±24.0 | 16.0±2.0  | 1.391±0.993        | 0.185 | –     |
| gradient-tight      | 100% | 0%   | 0%   | 85.0±24.0 | 16.0±2.0  | 1.351±0.943        | 0.185 | –     |
| post_proc           | 100% | 100% | 100% | 75.5±13.5 | 0.0±0.0   | 0.000±0.000        | 0.254 | –     |
| post_proc-tight     | 100% | 100% | 100% | 77.0±13.0 | 0.0±0.0   | 0.000±0.000        | 0.347 | –     |
| model_free          | 100% | 0%   | 0%   | 86.0±24.0 | 25.5±12.5 | 3.278±2.909        | 0.234 | –     |
| model_free-tight    | 100% | 0%   | 0%   | 86.0±24.0 | 25.5±11.5 | 2.773±2.399        | 0.239 | –     |
| dpcc-c-tight-dt0.25 | 100% | 100% | 100% | 76.0±2.0  | 0.0±0.0   | 0.000±0.000        | 0.529 | –     |



|                    |      |      |      |           |         |             |       |   |
|--------------------|------|------|------|-----------|---------|-------------|-------|---|
| dpcc-c-tight-dt0.5 | 100% | 100% | 100% | 68.5±2.5  | 0.0±0.0 | 0.000±0.000 | 0.329 | – |
| dpcc-c-tight-dt2.0 | 100% | 100% | 100% | 67.0±3.0  | 0.0±0.0 | 0.000±0.000 | 0.268 | – |
| dpcc-c-tight-dt4.0 | 100% | 100% | 100% | 124.5±4.5 | 0.0±0.0 | 0.000±0.000 | 0.254 | – |

#### F: FMV2 ODE=10 discrete-t — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std  | Viol#±std | Viol $\Sigma$ ±std | Time  | TE    |
|---------------------|------|------|------|------------|-----------|--------------------|-------|-------|
| dpcc-r              | 100% | 50%  | 50%  | 74.0±18.0  | 0.5±0.5   | 0.000±0.000        | 0.302 | –     |
| dpcc-r-tight        | 100% | 100% | 100% | 77.5±13.5  | 0.0±0.0   | 0.000±0.000        | 0.368 | –     |
| dpcc-c              | 100% | 100% | 100% | 63.5±6.5   | 0.0±0.0   | 0.000±0.000        | 0.337 | –     |
| dpcc-c-tight        | 100% | 100% | 100% | 67.0±7.0   | 0.0±0.0   | 0.000±0.000        | 0.344 | –     |
| dpcc-t              | 100% | 100% | 100% | 66.0±8.0   | 0.0±0.0   | 0.000±0.000        | 0.263 | –     |
| dpcc-t-tight        | 100% | 100% | 100% | 65.5±7.5   | 0.0±0.0   | 0.000±0.000        | 0.278 | –     |
| diffuser            | 100% | 0%   | 0%   | 65.5±8.5   | 10.0±1.0  | 0.181±0.022        | 0.170 | 0.029 |
| gradient            | 100% | 0%   | 0%   | 65.5±8.5   | 10.0±1.0  | 0.165±0.018        | 0.177 | –     |
| gradient-tight      | 100% | 0%   | 0%   | 65.5±8.5   | 10.0±1.0  | 0.162±0.021        | 0.181 | –     |
| post_proc           | 100% | 50%  | 50%  | 74.0±18.0  | 0.5±0.5   | 0.000±0.000        | 0.300 | –     |
| post_proc-tight     | 100% | 100% | 100% | 77.5±13.5  | 0.0±0.0   | 0.000±0.000        | 0.361 | –     |
| model_free          | 100% | 0%   | 0%   | 66.5±8.5   | 10.0±1.0  | 0.146±0.026        | 0.234 | –     |
| model_free-tight    | 100% | 0%   | 0%   | 66.0±9.0   | 10.0±1.0  | 0.143±0.028        | 0.249 | –     |
| dpcc-c-tight-dt0.25 | 100% | 50%  | 50%  | 137.5±37.5 | 51.5±51.5 | 0.840±0.840        | 2.577 | –     |
| dpcc-c-tight-dt0.5  | 100% | 100% | 100% | 76.0±9.0   | 0.0±0.0   | 0.001±0.000        | 0.523 | –     |
| dpcc-c-tight-dt2.0  | 100% | 100% | 100% | 69.5±3.5   | 0.0±0.0   | 0.000±0.000        | 0.267 | –     |
| dpcc-c-tight-dt4.0  | 100% | 100% | 100% | 142.5±8.5  | 0.0±0.0   | 0.000±0.000        | 0.248 | –     |

#### A: Diffusion — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std | Viol#±std | Viol $\Sigma$ ±std | Time  | TE    |
|---------------------|------|------|------|-----------|-----------|--------------------|-------|-------|
| dpcc-r              | 100% | 100% | 100% | 64.5±3.5  | 0.0±0.0   | 0.000±0.000        | 0.784 | –     |
| dpcc-r-tight        | 100% | 100% | 100% | 65.5±4.5  | 0.0±0.0   | 0.000±0.000        | 0.998 | –     |
| dpcc-c              | 100% | 50%  | 50%  | 62.5±4.5  | 1.0±1.0   | 0.003±0.003        | 0.592 | –     |
| dpcc-c-tight        | 100% | 100% | 100% | 63.5±3.5  | 0.0±0.0   | 0.000±0.000        | 0.822 | –     |
| dpcc-t              | 100% | 50%  | 50%  | 61.5±3.5  | 1.5±1.5   | 0.004±0.004        | 0.829 | –     |
| dpcc-t-tight        | 100% | 100% | 100% | 60.0±4.0  | 0.0±0.0   | 0.000±0.000        | 0.976 | –     |
| diffuser            | 100% | 0%   | 0%   | 59.0±3.0  | 20.0±2.0  | 2.515±0.387        | 0.306 | 0.025 |
| gradient            | 100% | 0%   | 0%   | 58.5±3.5  | 17.5±3.5  | 2.116±0.571        | 0.334 | –     |
| gradient-tight      | 100% | 0%   | 0%   | 59.0±3.0  | 19.5±1.5  | 2.218±0.316        | 0.348 | –     |
| post_proc           | 100% | 100% | 100% | 64.5±3.5  | 0.0±0.0   | 0.000±0.000        | 0.782 | –     |
| post_proc-tight     | 100% | 100% | 100% | 65.5±4.5  | 0.0±0.0   | 0.000±0.000        | 0.996 | –     |
| model_free          | 100% | 0%   | 0%   | 63.5±2.5  | 19.0±0.0  | 2.829±0.455        | 0.458 | –     |
| model_free-tight    | 100% | 0%   | 0%   | 64.5±3.5  | 15.0±2.0  | 2.029±0.233        | 0.470 | –     |
| dpcc-c-tight-dt0.25 | 100% | 100% | 100% | 70.0±3.0  | 0.0±0.0   | 0.000±0.000        | 1.259 | –     |
| dpcc-c-tight-dt0.5  | 100% | 100% | 100% | 63.5±0.5  | 0.0±0.0   | 0.000±0.000        | 0.937 | –     |

|                    |      |      |      |           |          |             |       |   |
|--------------------|------|------|------|-----------|----------|-------------|-------|---|
| dpcc-c-tight-dt2.0 | 100% | 100% | 100% | 64.5±0.5  | 0.0±0.0  | 0.000±0.000 | 0.679 | – |
| dpcc-c-tight-dt4.0 | 100% | 0%   | 0%   | 107.0±2.0 | 16.5±0.5 | 3.711±0.053 | 0.810 | – |

## 2.3 Both-Hard

### G: FMv3 continuous-t — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std  | Viol#±std | Viol $\Sigma$ ±std | Time  | TE    |
|---------------------|------|------|------|------------|-----------|--------------------|-------|-------|
| dpcc-r              | 100% | 50%  | 50%  | 92.0±32.0  | 35.5±35.5 | 0.478±0.478        | 0.331 | –     |
| dpcc-r-tight        | 100% | 100% | 100% | 57.5±0.5   | 0.0±0.0   | 0.000±0.000        | 0.337 | –     |
| dpcc-c              | 100% | 0%   | 0%   | 138.0±12.0 | 55.5±8.5  | 0.598±0.276        | 0.286 | –     |
| dpcc-c-tight        | 100% | 100% | 100% | 58.5±3.5   | 0.0±0.0   | 0.000±0.000        | 0.519 | –     |
| dpcc-t              | 100% | 0%   | 0%   | 135.0±4.0  | 65.5±0.5  | 0.736±0.074        | 0.294 | –     |
| dpcc-t-tight        | 100% | 100% | 100% | 64.0±1.0   | 0.0±0.0   | 0.000±0.000        | 0.398 | –     |
| diffuser            | 100% | 50%  | 50%  | 84.5±24.5  | 2.0±2.0   | 0.026±0.010        | 0.173 | 0.029 |
| gradient            | 100% | 50%  | 50%  | 84.5±23.5  | 2.0±2.0   | 0.023±0.005        | 0.187 | –     |
| gradient-tight      | 100% | 0%   | 0%   | 81.0±20.0  | 8.5±5.5   | 1.602±1.581        | 0.186 | –     |
| post_proc           | 100% | 50%  | 50%  | 92.0±32.0  | 35.5±35.5 | 0.478±0.478        | 0.335 | –     |
| post_proc-tight     | 100% | 100% | 100% | 57.5±0.5   | 0.0±0.0   | 0.000±0.000        | 0.331 | –     |
| model_free          | 100% | 50%  | 50%  | 85.0±23.0  | 2.0±2.0   | 0.006±0.006        | 0.239 | –     |
| model_free-tight    | 100% | 50%  | 50%  | 85.5±23.5  | 1.5±1.5   | 0.003±0.003        | 0.250 | –     |
| dpcc-c-tight-dt0.25 | 100% | 100% | 100% | 70.0±2.0   | 0.0±0.0   | 0.000±0.000        | 0.666 | –     |
| dpcc-c-tight-dt0.5  | 100% | 100% | 100% | 62.0±5.0   | 0.0±0.0   | 0.001±0.001        | 0.449 | –     |
| dpcc-c-tight-dt2.0  | 100% | 100% | 100% | 72.5±8.5   | 0.0±0.0   | 0.000±0.000        | 0.323 | –     |
| dpcc-c-tight-dt4.0  | 100% | 100% | 100% | 101.0±2.0  | 0.0±0.0   | 0.000±0.000        | 0.297 | –     |

### F: FMv2 ODE=10 discrete-t — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std  | Viol#±std | Viol $\Sigma$ ±std | Time  | TE    |
|---------------------|------|------|------|------------|-----------|--------------------|-------|-------|
| dpcc-r              | 100% | 50%  | 50%  | 97.5±34.5  | 36.5±36.5 | 0.388±0.388        | 0.303 | –     |
| dpcc-r-tight        | 100% | 100% | 100% | 62.0±2.0   | 0.0±0.0   | 0.000±0.000        | 0.346 | –     |
| dpcc-c              | 100% | 100% | 100% | 61.5±1.5   | 0.0±0.0   | 0.000±0.000        | 0.339 | –     |
| dpcc-c-tight        | 100% | 100% | 100% | 61.5±4.5   | 0.0±0.0   | 0.000±0.000        | 0.401 | –     |
| dpcc-t              | 100% | 0%   | 0%   | 103.0±44.0 | 35.5±32.5 | 0.294±0.283        | 0.322 | –     |
| dpcc-t-tight        | 100% | 100% | 100% | 57.0±1.0   | 0.0±0.0   | 0.000±0.000        | 0.318 | –     |
| diffuser            | 100% | 0%   | 0%   | 65.5±8.5   | 11.5±0.5  | 3.013±0.172        | 0.170 | 0.029 |
| gradient            | 100% | 0%   | 0%   | 65.5±8.5   | 11.5±0.5  | 2.804±0.074        | 0.187 | –     |
| gradient-tight      | 100% | 0%   | 0%   | 65.5±8.5   | 12.0±1.0  | 2.844±0.109        | 0.187 | –     |
| post_proc           | 100% | 50%  | 50%  | 97.5±34.5  | 36.5±36.5 | 0.388±0.388        | 0.300 | –     |
| post_proc-tight     | 100% | 100% | 100% | 62.0±2.0   | 0.0±0.0   | 0.000±0.000        | 0.345 | –     |
| model_free          | 100% | 50%  | 50%  | 66.5±8.5   | 6.0±6.0   | 1.445±1.445        | 0.240 | –     |
| model_free-tight    | 100% | 50%  | 50%  | 66.5±8.5   | 6.0±6.0   | 1.430±1.430        | 0.240 | –     |
| dpcc-c-tight-dt0.25 | 100% | 100% | 100% | 71.5±3.5   | 0.0±0.0   | 0.000±0.000        | 0.572 | –     |
| dpcc-c-tight-dt0.5  | 100% | 100% | 100% | 60.5±5.5   | 0.0±0.0   | 0.000±0.000        | 0.556 | –     |

|                    |      |      |      |           |         |             |       |   |
|--------------------|------|------|------|-----------|---------|-------------|-------|---|
| dpcc-c-tight-dt2.0 | 100% | 100% | 100% | 72.0±14.0 | 0.0±0.0 | 0.000±0.000 | 0.319 | – |
| dpcc-c-tight-dt4.0 | 100% | 100% | 100% | 96.0±20.0 | 0.0±0.0 | 0.000±0.000 | 0.315 | – |

#### A: Diffusion — Seed 6

| Method              | SR   | CS   | G&C; | Steps±std | Viol#±std | Viol $\Sigma$ ±std | Time  | TE    |
|---------------------|------|------|------|-----------|-----------|--------------------|-------|-------|
| dpcc-r              | 100% | 0%   | 0%   | 61.5±0.5  | 2.5±0.5   | 0.002±0.001        | 1.019 | –     |
| dpcc-r-tight        | 100% | 100% | 100% | 66.0±0.0  | 0.0±0.0   | 0.000±0.000        | 0.948 | –     |
| dpcc-c              | 100% | 0%   | 0%   | 84.5±11.5 | 19.0±17.0 | 0.381±0.377        | 0.704 | –     |
| dpcc-c-tight        | 100% | 100% | 100% | 56.5±2.5  | 0.0±0.0   | 0.000±0.000        | 0.997 | –     |
| dpcc-t              | 100% | 50%  | 50%  | 81.5±18.5 | 17.0±17.0 | 0.416±0.416        | 0.697 | –     |
| dpcc-t-tight        | 100% | 100% | 100% | 62.5±2.5  | 0.0±0.0   | 0.000±0.000        | 0.964 | –     |
| diffuser            | 100% | 0%   | 0%   | 59.0±3.0  | 7.5±0.5   | 0.090±0.001        | 0.307 | 0.025 |
| gradient            | 100% | 0%   | 0%   | 59.0±3.0  | 7.0±1.0   | 0.064±0.007        | 0.331 | –     |
| gradient-tight      | 100% | 0%   | 0%   | 61.0±3.0  | 6.0±1.0   | 0.044±0.004        | 0.332 | –     |
| post_proc           | 100% | 0%   | 0%   | 61.5±0.5  | 2.5±0.5   | 0.002±0.001        | 1.029 | –     |
| post_proc-tight     | 100% | 100% | 100% | 66.0±0.0  | 0.0±0.0   | 0.000±0.000        | 0.924 | –     |
| model_free          | 100% | 0%   | 0%   | 63.0±2.0  | 8.5±0.5   | 0.103±0.008        | 0.465 | –     |
| model_free-tight    | 100% | 0%   | 0%   | 63.5±2.5  | 8.5±0.5   | 0.091±0.010        | 0.481 | –     |
| dpcc-c-tight-dt0.25 | 100% | 50%  | 50%  | 75.5±0.5  | 0.5±0.5   | 0.000±0.000        | 1.835 | –     |
| dpcc-c-tight-dt0.5  | 100% | 100% | 100% | 68.5±4.5  | 0.0±0.0   | 0.000±0.000        | 1.185 | –     |
| dpcc-c-tight-dt2.0  | 100% | 100% | 100% | 64.5±2.5  | 0.0±0.0   | 0.000±0.000        | 0.737 | –     |
| dpcc-c-tight-dt4.0  | 100% | 100% | 100% | 85.0±2.0  | 0.0±0.0   | 0.000±0.000        | 0.691 | –     |

## Section 3 — Three-Way Head-to-Head Comparison

Best G&C; highlighted green. Best (lowest) violation count highlighted teal.  $\Delta GF$  = G minus F G&C; score, showing the continuous-t upgrade effect per method.

### Top-Right-Hard

| Method              | A<br>G&C; | F<br>G&C; | G<br>G&C; | Best  | $\Delta GF$ | A<br>Viol | F<br>Viol | G<br>Viol | F Time | G Time |
|---------------------|-----------|-----------|-----------|-------|-------------|-----------|-----------|-----------|--------|--------|
| dpcc-r              | 100%      | 0%        | 50%       | A     | +50p<br>p   | 0.0       | 1.5       | 4.5       | 0.222  | 0.275  |
| dpcc-r-tight        | 100%      | 100%      | 100%      | A/F/G | =           | 0.0       | 0.0       | 0.0       | 0.283  | 0.318  |
| dpcc-c              | 100%      | 50%       | 0%        | A     | -50pp       | 0.0       | 1.0       | 1.5       | 0.224  | 0.253  |
| dpcc-c-tight        | 100%      | 100%      | 100%      | A/F/G | =           | 0.0       | 0.0       | 0.0       | 0.243  | 0.349  |
| dpcc-t              | 50%       | 50%       | 0%        | A/F   | -50pp       | 0.5       | 0.0       | 2.5       | 0.239  | 0.275  |
| dpcc-t-tight        | 100%      | 100%      | 100%      | A/F/G | =           | 0.0       | 0.0       | 0.0       | 0.267  | 0.314  |
| diffuser            | 100%      | 0%        | 0%        | A     | =           | 0.0       | 33.0      | 23.5      | 0.164  | 0.175  |
| gradient            | 100%      | 0%        | 50%       | A     | +50p<br>p   | 0.0       | 34.0      | 16.0      | 0.175  | 0.175  |
| gradient-tight      | 0%        | 0%        | 50%       | G     | +50p<br>p   | 15.0      | 34.0      | 16.0      | 0.175  | 0.180  |
| post_proc           | 100%      | 0%        | 50%       | A     | +50p<br>p   | 0.0       | 1.5       | 4.5       | 0.213  | 0.265  |
| post_proc-tight     | 100%      | 100%      | 100%      | A/F/G | =           | 0.0       | 0.0       | 0.0       | 0.283  | 0.321  |
| model_free          | 0%        | 0%        | 0%        | A/F/G | =           | 3.0       | 35.5      | 24.5      | 0.214  | 0.224  |
| model_free-tight    | 50%       | 0%        | 0%        | A     | =           | 2.0       | 36.0      | 24.5      | 0.233  | 0.232  |
| dpcc-c-tight-dt0.25 | 100%      | 50%       | 100%      | A/G   | +50p<br>p   | 0.0       | 1.0       | 0.0       | 0.452  | 0.681  |
| dpcc-c-tight-dt0.5  | 0%        | 100%      | 100%      | F/G   | =           | 1.0       | 0.0       | 0.0       | 0.275  | 0.426  |
| dpcc-c-tight-dt2.0  | 100%      | 100%      | 100%      | A/F/G | =           | 0.0       | 0.0       | 0.0       | 0.248  | 0.272  |
| dpcc-c-tight-dt4.0  | 50%       | 100%      | 100%      | F/G   | =           | 0.0       | 0.0       | 0.0       | 0.260  | 0.270  |

### Top-Left-Hard

| Method         | A<br>G&C; | F<br>G&C; | G<br>G&C; | Best  | $\Delta GF$ | A<br>Viol | F<br>Viol | G<br>Viol | F Time | G Time |
|----------------|-----------|-----------|-----------|-------|-------------|-----------|-----------|-----------|--------|--------|
| dpcc-r         | 100%      | 50%       | 100%      | A/G   | +50p<br>p   | 0.0       | 0.5       | 0.0       | 0.302  | 0.262  |
| dpcc-r-tight   | 100%      | 100%      | 100%      | A/F/G | =           | 0.0       | 0.0       | 0.0       | 0.368  | 0.346  |
| dpcc-c         | 50%       | 100%      | 100%      | F/G   | =           | 1.0       | 0.0       | 0.0       | 0.337  | 0.258  |
| dpcc-c-tight   | 100%      | 100%      | 100%      | A/F/G | =           | 0.0       | 0.0       | 0.0       | 0.344  | 0.282  |
| dpcc-t         | 50%       | 100%      | 50%       | F     | -50pp       | 1.5       | 0.0       | 22.0      | 0.263  | 0.257  |
| dpcc-t-tight   | 100%      | 100%      | 100%      | A/F/G | =           | 0.0       | 0.0       | 0.0       | 0.278  | 0.259  |
| diffuser       | 0%        | 0%        | 0%        | A/F/G | =           | 20.0      | 10.0      | 24.5      | 0.170  | 0.173  |
| gradient       | 0%        | 0%        | 0%        | A/F/G | =           | 17.5      | 10.0      | 16.0      | 0.177  | 0.185  |
| gradient-tight | 0%        | 0%        | 0%        | A/F/G | =           | 19.5      | 10.0      | 16.0      | 0.181  | 0.185  |

|                     |      |      |      |       |           |      |      |      |       |       |
|---------------------|------|------|------|-------|-----------|------|------|------|-------|-------|
| post_proc           | 100% | 50%  | 100% | A/G   | +50p<br>p | 0.0  | 0.5  | 0.0  | 0.300 | 0.254 |
| post_proc-tight     | 100% | 100% | 100% | A/F/G | =         | 0.0  | 0.0  | 0.0  | 0.361 | 0.347 |
| model_free          | 0%   | 0%   | 0%   | A/F/G | =         | 19.0 | 10.0 | 25.5 | 0.234 | 0.234 |
| model_free-tight    | 0%   | 0%   | 0%   | A/F/G | =         | 15.0 | 10.0 | 25.5 | 0.249 | 0.239 |
| dpcc-c-tight-dt0.25 | 100% | 50%  | 100% | A/G   | +50p<br>p | 0.0  | 51.5 | 0.0  | 2.577 | 0.529 |
| dpcc-c-tight-dt0.5  | 100% | 100% | 100% | A/F/G | =         | 0.0  | 0.0  | 0.0  | 0.523 | 0.329 |
| dpcc-c-tight-dt2.0  | 100% | 100% | 100% | A/F/G | =         | 0.0  | 0.0  | 0.0  | 0.267 | 0.268 |
| dpcc-c-tight-dt4.0  | 0%   | 100% | 100% | F/G   | =         | 16.5 | 0.0  | 0.0  | 0.248 | 0.254 |

## Both-Hard

| Method              | A<br>G&C; | F<br>G&C; | G<br>G&C; | Best  | ΔGF        | A<br>Viol | F<br>Viol | G<br>Viol | F Time | G Time |
|---------------------|-----------|-----------|-----------|-------|------------|-----------|-----------|-----------|--------|--------|
| dpcc-r              | 0%        | 50%       | 50%       | F/G   | =          | 2.5       | 36.5      | 35.5      | 0.303  | 0.331  |
| dpcc-r-tight        | 100%      | 100%      | 100%      | A/F/G | =          | 0.0       | 0.0       | 0.0       | 0.346  | 0.337  |
| dpcc-c              | 0%        | 100%      | 0%        | F     | -100p<br>p | 19.0      | 0.0       | 55.5      | 0.339  | 0.286  |
| dpcc-c-tight        | 100%      | 100%      | 100%      | A/F/G | =          | 0.0       | 0.0       | 0.0       | 0.401  | 0.519  |
| dpcc-t              | 50%       | 0%        | 0%        | A     | =          | 17.0      | 35.5      | 65.5      | 0.322  | 0.294  |
| dpcc-t-tight        | 100%      | 100%      | 100%      | A/F/G | =          | 0.0       | 0.0       | 0.0       | 0.318  | 0.398  |
| diffuser            | 0%        | 0%        | 50%       | G     | +50p<br>p  | 7.5       | 11.5      | 2.0       | 0.170  | 0.173  |
| gradient            | 0%        | 0%        | 50%       | G     | +50p<br>p  | 7.0       | 11.5      | 2.0       | 0.187  | 0.187  |
| gradient-tight      | 0%        | 0%        | 0%        | A/F/G | =          | 6.0       | 12.0      | 8.5       | 0.187  | 0.186  |
| post_proc           | 0%        | 50%       | 50%       | F/G   | =          | 2.5       | 36.5      | 35.5      | 0.300  | 0.335  |
| post_proc-tight     | 100%      | 100%      | 100%      | A/F/G | =          | 0.0       | 0.0       | 0.0       | 0.345  | 0.331  |
| model_free          | 0%        | 50%       | 50%       | F/G   | =          | 8.5       | 6.0       | 2.0       | 0.240  | 0.239  |
| model_free-tight    | 0%        | 50%       | 50%       | F/G   | =          | 8.5       | 6.0       | 1.5       | 0.240  | 0.250  |
| dpcc-c-tight-dt0.25 | 50%       | 100%      | 100%      | F/G   | =          | 0.5       | 0.0       | 0.0       | 0.572  | 0.666  |
| dpcc-c-tight-dt0.5  | 100%      | 100%      | 100%      | A/F/G | =          | 0.0       | 0.0       | 0.0       | 0.556  | 0.449  |
| dpcc-c-tight-dt2.0  | 100%      | 100%      | 100%      | A/F/G | =          | 0.0       | 0.0       | 0.0       | 0.319  | 0.323  |
| dpcc-c-tight-dt4.0  | 100%      | 100%      | 100%      | A/F/G | =          | 0.0       | 0.0       | 0.0       | 0.315  | 0.297  |

## Section 4 — dt Sensitivity: Complete Data

G resolves F's pathological dt=0.25 instability on top-left-hard. Both G and F achieve 100% on all other dt values across all scenarios for dpcc-c-tightened.

### Top-Right-Hard

| dt           | A<br>G&C; | F<br>G&C; | G<br>G&C; | $\Delta$ GF | A Steps $\pm$ std | G Steps $\pm$ std | A Time | F Time | G Time |
|--------------|-----------|-----------|-----------|-------------|-------------------|-------------------|--------|--------|--------|
| dt=0.25      | 100%      | 50%       | 100%      | +50p<br>p   | 81.0 $\pm$ 6.0    | 78.5 $\pm$ 3.5    | 2.225  | 0.452  | 0.681  |
| dt=0.5       | 0%        | 100%      | 100%      | =           | 75.5 $\pm$ 2.5    | 77.5 $\pm$ 5.5    | 1.668  | 0.275  | 0.426  |
| dt=1.0 (def) | 100%      | 100%      | 100%      | =           | 66.5 $\pm$ 0.5    | 71.0 $\pm$ 5.0    | 0.911  | 0.243  | 0.349  |
| dt=2.0       | 100%      | 100%      | 100%      | =           | 66.5 $\pm$ 0.5    | 71.0 $\pm$ 6.0    | 0.689  | 0.248  | 0.272  |
| dt=4.0       | 50%       | 100%      | 100%      | =           | 72.0 $\pm$ 0.0    | 131.5 $\pm$ 1.5   | 0.535  | 0.260  | 0.270  |

### Top-Left-Hard

| dt           | A<br>G&C; | F<br>G&C; | G<br>G&C; | $\Delta$ GF | A Steps $\pm$ std | G Steps $\pm$ std | A Time | F Time | G Time |
|--------------|-----------|-----------|-----------|-------------|-------------------|-------------------|--------|--------|--------|
| dt=0.25      | 100%      | 50%       | 100%      | +50p<br>p   | 70.0 $\pm$ 3.0    | 76.0 $\pm$ 2.0    | 1.259  | 2.577  | 0.529  |
| dt=0.5       | 100%      | 100%      | 100%      | =           | 63.5 $\pm$ 0.5    | 68.5 $\pm$ 2.5    | 0.937  | 0.523  | 0.329  |
| dt=1.0 (def) | 100%      | 100%      | 100%      | =           | 63.5 $\pm$ 3.5    | 67.0 $\pm$ 2.0    | 0.822  | 0.344  | 0.282  |
| dt=2.0       | 100%      | 100%      | 100%      | =           | 64.5 $\pm$ 0.5    | 67.0 $\pm$ 3.0    | 0.679  | 0.267  | 0.268  |
| dt=4.0       | 0%        | 100%      | 100%      | =           | 107.0 $\pm$ 2.0   | 124.5 $\pm$ 4.5   | 0.810  | 0.248  | 0.254  |

### Both-Hard

| dt           | A<br>G&C; | F<br>G&C; | G<br>G&C; | $\Delta$ GF | A Steps $\pm$ std | G Steps $\pm$ std | A Time | F Time | G Time |
|--------------|-----------|-----------|-----------|-------------|-------------------|-------------------|--------|--------|--------|
| dt=0.25      | 50%       | 100%      | 100%      | =           | 75.5 $\pm$ 0.5    | 70.0 $\pm$ 2.0    | 1.835  | 0.572  | 0.666  |
| dt=0.5       | 100%      | 100%      | 100%      | =           | 68.5 $\pm$ 4.5    | 62.0 $\pm$ 5.0    | 1.185  | 0.556  | 0.449  |
| dt=1.0 (def) | 100%      | 100%      | 100%      | =           | 56.5 $\pm$ 2.5    | 58.5 $\pm$ 3.5    | 0.997  | 0.401  | 0.519  |
| dt=2.0       | 100%      | 100%      | 100%      | =           | 64.5 $\pm$ 2.5    | 72.5 $\pm$ 8.5    | 0.737  | 0.319  | 0.323  |
| dt=4.0       | 100%      | 100%      | 100%      | =           | 85.0 $\pm$ 2.0    | 101.0 $\pm$ 2.0   | 0.691  | 0.315  | 0.297  |

## Section 5 — Summary Statistics & Architecture Analysis

### 5.1 Aggregate Statistics — Core Methods (13×3=39 observations each)

| Metric            | A (Diffusion) | F (FMv2 ODE=10) | G (FMv3 cont-t) | G-F $\Delta$ | G-A $\Delta$ |
|-------------------|---------------|-----------------|-----------------|--------------|--------------|
| G&C; Success      | 0.551         | 0.487           | 0.526           | +0.038       | -0.026       |
| Constraint Sat.   | 0.551         | 0.487           | 0.526           | +0.038       | -0.026       |
| Task Success      | 0.987         | 0.923           | 0.974           | +0.051       | -0.013       |
| Avg Violations    | 4.936         | 9.821           | 11.667          | +1.846       | +6.731       |
| Avg Total Viol.   | 0.372         | 1.398           | 0.722           | -0.675       | +0.351       |
| Avg Time (s/step) | 0.671         | 0.261           | 0.269           | +0.008       | -0.401       |

### 5.2 Win Counts — G&C; (39 comparisons)

|      |      |      |          |
|------|------|------|----------|
| A: 7 | F: 2 | G: 3 | Ties: 27 |
|------|------|------|----------|

### 5.3 Key Findings: FMv3 Continuous-t Architecture

#### 1. G (FMv3) substantially improves both-hard — the most important gain

On both-hard, G achieves 50% G&C; on diffuser, gradient, model\_free, and model\_free-tight — methods that score 0% for F. This is the most significant quality improvement from continuous-t. G also achieves 100% on all tightened DPCC methods on both-hard, matching F.

#### 2. G improves top-right-hard on untightened DPCC and baseline methods

dpcc-r goes from 0% (F) to 50% (G) on top-right-hard. gradient/gradient-tight go from 0% (F) to 50% (G). post\_proc goes from 0% (F) to 50% (G). Continuous-t conditioning makes the model more robust to constraint-challenging trajectories even without tightening.

#### 3. G resolves F's catastrophic dt=0.25 instability on top-left-hard

F at dt=0.25 on top-left-hard produces 51.5 avg violations over 137 steps — a pathological failure. G resolves this completely: 0 violations, 76 steps, 100% G&C.; This confirms that continuous-t conditioning improves ODE integration stability at small step sizes.

#### 4. G regresses on top-right dpcc-c and dpcc-t (untightened)

dpcc-c goes from 50% (F, clean trajectory) to 0% (G) on top-right-hard. dpcc-t goes from 50% (F) to 0% (G). These failures introduce violations (1.5–2.5 avg) where F had none. The continuous-t parameterisation changes the trajectory shape in ways that occasionally miss the goal on this specific scenario without tightening.

#### 5. Tightened DPCC methods are fully reliable for G across all scenarios

dpcc-r-tight, dpcc-c-tight, dpcc-t-tight, and post\_proc-tight all achieve 100% G&C; for G on all three scenarios. These remain the safest deployment configuration.

#### 6. G generates longer trajectories — a consequence of continuous-t dynamics

G's avg steps are notably higher on several methods (dpcc-r: 82 vs F's 0/75, dpcc-c: 138 steps on both-hard, dpcc-t: 135 on both-hard). Continuous-t ODE integration produces smoother but longer paths. Tightened variants remain efficient (57–64 avg steps on both-hard).

#### 7. G's computation time is slightly higher than F due to continuous-t evaluation

G averages 0.269s/step vs F's 0.261s/step — a 3.0% overhead. Both remain well below A's 0.671s/step for DPCC methods. The overhead from querying continuous-t is modest and dominated by the projection step.

#### 8. Overall verdict: FMv3 (G) is the strongest FM model tested, approaching Diffusion

G wins 3 comparisons vs F=2, A=7 (ties=27). The continuous-t upgrade meaningfully improves robustness on both-hard and top-left-hard. The recommended configuration remains G with tightened DPCC methods (dpcc-c-tight or dpcc-t-tight) at  $dt=0.5-2.0$ , where G achieves 100% G&C; with low violations and competitive speed.

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*A=Diffusion (diffusion/6, Seed 6). F=FMv2 (flow\_matching\_v2/6, aw=1, ODE steps=10, discrete-t bins, ckpt 90000). G=FMv3 (flow\_matching\_v3/6, aw=1, ODE steps=10, continuous real-valued  $t \in [0,1]$ , ckpt 90000). All Seed 6, avoiding-d3il, H8\_K20\_Dmodels.*